

Task 2 Report: CNN Baselines and the First CNN+CBAM Model

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1. Experimental question

The first modeling stage asks whether channel-spatial attention improves a custom CNN on the SOMLAP feature grids. The decisive comparison is between the plain CNN and CNN+CBAM because they share the same convolutional backbone and classification head. VGG16, ResNet50 and EfficientNetB0 broaden the baseline set and show whether substantially larger image architectures offer a useful advantage on this synthetic representation.

Model quality cannot be summarized by accuracy alone. With fewer malware than benign records, a threshold may preserve overall accuracy while either missing malware or producing excessive false alarms. Macro-F1 is therefore the primary selection measure; malware precision, recall and F1 describe the operating trade-off, and ROC-AUC and PR-AUC evaluate ranking across thresholds.

2. Split and leakage controls

The 108 predictors are hashed as one vector before partitioning. Records with the same hash form an indivisible group. The resulting split contains 36,027 training samples, 7,651 validation samples and 7,730 test samples, corresponding to 25,928, 5,556 and 5,557 distinct groups. The test partition includes 4,753 benign and 2,977 malware records. An explicit overlap check confirms that no exact-vector group appears in more than one partition.

Partition	Samples	Unique groups	Benign	Malware	Malware share
Training	36,027	25,928	22,108	13,919	38.63%
Validation	7,651	5,556	4,739	2,912	38.06%
Testing	7,730	5,557	4,753	2,977	38.51%

Table 1. Exact-duplicate-group split used by every evaluated model.

This grouping rule blocks the most direct form of leakage found during EDA. It does not establish independence by malware family, original executable, collection host or time because those identifiers are absent from the CSV. Comparisons with published SOMLAP scores must therefore account for the fact that the current split is exact-duplicate-safe, not source-independent.

3. Preprocessing and training settings

Standardization and intensity mapping are fitted only on the training partition. Each transformed vector is reshaped to 12×9 , resized to 96×96 , and repeated across three channels. Gaussian intensity noise with standard deviation 1.5 is applied during training; validation and test inputs remain deterministic. Class weights compensate for the 61.5:38.5 class distribution.

Training uses TensorFlow MirroredStrategy on two NVIDIA T4 GPUs with a per-replica batch size of 16 and a global batch size of 32. Every fit call has a 60-epoch ceiling. Early stopping restores the weights with the best

validation loss, while ReduceLROnPlateau lowers the learning rate after a plateau. The first CNN+CBAM run stopped after 23 epochs, restored epoch 13, and required 1,071.9 seconds.

4. Baseline models

4.1 Plain CNN

The plain CNN is the matched no-attention baseline. Four convolution blocks increase channel width from 32 to 256; each convolution is followed by batch normalization, and the first three blocks end in max pooling. Global average pooling feeds dense layers of 256 and 128 units and a sigmoid output. The model has 489,825 parameters and a serialized size of 5.70 MB.

4.2 VGG16

VGG16 represents a deep stack of uniform convolutions. With the classification head used here, it has 14.85 million parameters and occupies 57.73 MB. Its inclusion tests whether a conventional high-capacity feature extractor improves the fixed-grid task; its held-out behavior ultimately favors malware recall at the cost of many benign false positives.

4.3 ResNet50

ResNet50 uses residual connections to optimize a substantially deeper feature extractor. It is the largest model in the experiment, with 24.11 million parameters and a 96.62 MB file. This baseline tests whether depth and residual learning compensate for the artificial neighborhood structure of the input.

4.4 EfficientNetB0

EfficientNetB0 combines depthwise operations with compound scaling and contains 4.05 million parameters. Although smaller than VGG16 and ResNet50, it remains considerably larger than the custom networks. At the fixed threshold it is the most conservative malware predictor, yielding high precision and low recall.

5. Proposed CNN with CBAM

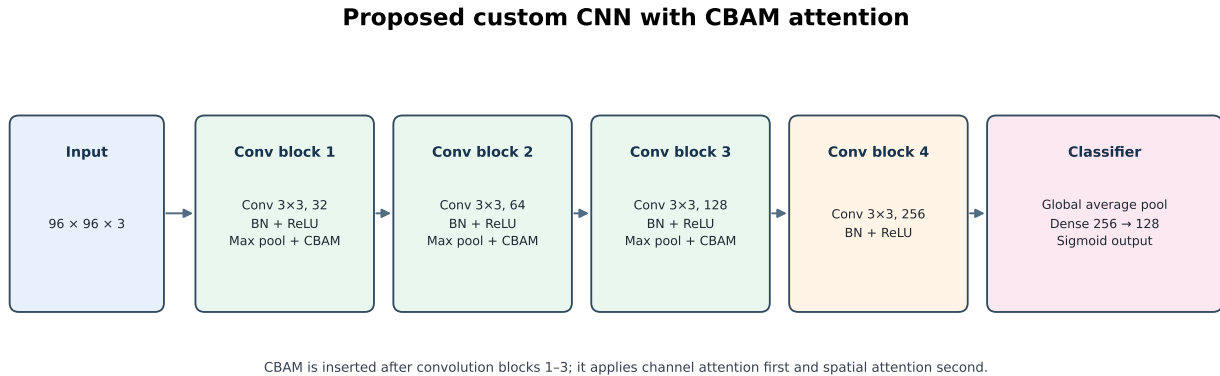


Figure 1: Custom four-block CNN with CBAM after the first three convolution blocks.

CBAM applies channel attention followed by spatial attention [8]. The channel branch summarizes every feature map with global average and maximum pooling. A shared two-layer bottleneck with reduction ratio 16 converts those summaries into sigmoid weights, which amplify or suppress channels for the current input.

The spatial branch computes average and maximum projections across the weighted channels, concatenates them, and applies a 7×7 convolution followed by a sigmoid. The resulting mask reweights positions in the convolutional

tensor. Because those positions originate from a resized feature grid, they represent learned combinations of PE fields rather than physical executable regions.

CBAM is inserted after the first three convolution blocks. The backbone, normalization layers, dense head, dropout settings and sigmoid output otherwise match the plain CNN. Attention adds 2,982 parameters, producing a total of 492,807. The small parameter difference makes the custom pair a more controlled test of attention than comparisons against the transfer-learning networks.

6. Evaluation measures

Accuracy reports the overall fraction of correct decisions. Malware precision measures the proportion of malware predictions that are correct, whereas recall measures the proportion of malware records detected. Their harmonic mean gives malware F1. Macro-F1 gives equal weight to benign and malware F1; weighted F1 weights each class by its support. ROC-AUC and PR-AUC summarize ranking across thresholds, with PR-AUC placing greater emphasis on the positive-class precision-recall trade-off.

The evaluation also retains per-class support and the four confusion-matrix counts so that aggregate values can be checked against actual errors. Training time, inference time, parameter count and serialized size describe computation and storage. Every curve has labelled axes, a legend and a no-skill reference where appropriate.

7. Held-out results

Model	Acc.	Prec.	Recall	F1	Macro-F1	Weighted F1	ROC-AUC	PR-AUC	Train s	Parameters	MB
Plain CNN	0.9508	0.9315	0.9416	0.9365	0.9482	0.9509	0.9931	0.9870	1213.3	489,825	5.70
VGG16	0.9009	0.7973	0.9960	0.8856	0.8991	0.9022	0.9950	0.9925	520.0	14,846,273	57.73
ResNet50	0.9247	0.8544	0.9698	0.9084	0.9223	0.9254	0.9875	0.9822	458.8	24,112,513	96.62
EfficientNetB0	0.8492	0.9773	0.6228	0.7608	0.8253	0.8401	0.9670	0.9486	484.9	4,050,852	16.27
First CNN+CBAM	0.9492	0.8906	0.9896	0.9375	0.9473	0.9496	0.9954	0.9910	1071.9	492,807	5.78

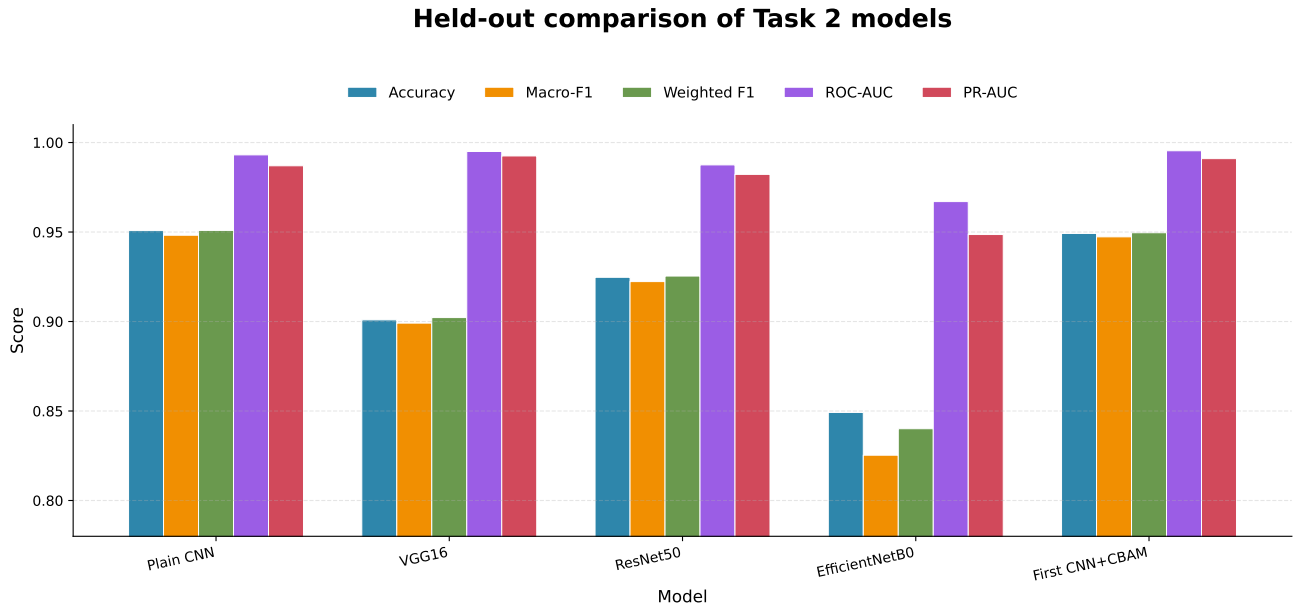


Figure 2: Held-out comparison of accuracy, Macro-F1, weighted F1, ROC-AUC and PR-AUC.

7.1 Plain CNN

The plain CNN achieves 0.9508 accuracy and 0.9482 Macro-F1, the strongest balanced performance among the four baselines. Malware precision is 0.9315 and recall 0.9416. Its advantage is not merely computational: at threshold 0.5 it also maintains a better balance between the two malware error types than the larger baselines.

7.2 VGG16

VGG16 detects 99.60% of malware records, but malware precision falls to 0.7973 because many benign files are flagged. ROC-AUC 0.9950 and PR-AUC 0.9925 show that the underlying scores still rank the classes well. Its lower Macro-F1 is therefore tied mainly to the fixed operating threshold rather than a complete loss of separability.

7.3 ResNet50

ResNet50 yields 0.9223 Macro-F1 and 0.9698 malware recall. These values improve the precision-recall balance relative to VGG16 but remain below the plain CNN. The result, together with the highest parameter count and file size, shows that greater depth does not provide a corresponding gain on the feature grids.

7.4 EfficientNetB0

EfficientNetB0 reaches malware precision of 0.9773 but recall of only 0.6228, meaning that more than one third of malware records are missed. Accuracy 0.8492 and Macro-F1 0.8253 expose the weakness that would be hidden by reporting precision alone.

7.5 First CNN+CBAM result

The first CNN+CBAM model records 0.9492 accuracy and 0.9473 Macro-F1. Malware recall rises to 0.9896, while precision falls to 0.8906. The corresponding confusion matrix contains 31 malware misses and 362 benign false alarms. Against the matched plain CNN, Macro-F1 is lower by 0.0009 and accuracy by 0.0017. The first run therefore changes the error profile without improving the primary metric.

Class	Precision	Recall	F1	Support
Benign	0.9930	0.9238	0.9572	4,753
Malware	0.8906	0.9896	0.9375	2,977
Macro average	0.9418	0.9567	0.9473	7,730
Weighted average	0.9535	0.9492	0.9496	7,730

The attention model records ROC-AUC 0.9954 and PR-AUC 0.9910. Those values indicate strong score ranking over many thresholds, but they do not remove the precision loss observed at 0.5. A different threshold could exchange false positives for false negatives; selecting it after viewing the test set would bias the estimate, so no post-hoc threshold adjustment is reported.

8. Effect of attention

The first matched comparison provides no evidence of a Macro-F1 gain from attention. CBAM raises malware recall from 0.9416 to 0.9896 but reduces precision from 0.9315 to 0.8906. Whether that operating point is preferable depends on the relative cost of a missed infection and a false alarm; the experiment itself does not assign those costs.

Both custom models in Task 2 include dropout. The comparison therefore does not show whether the observed difference is stable when regularization changes. Task 3 resolves this ambiguity through a two-by-two design containing plain and CBAM networks with dropout both enabled and disabled.

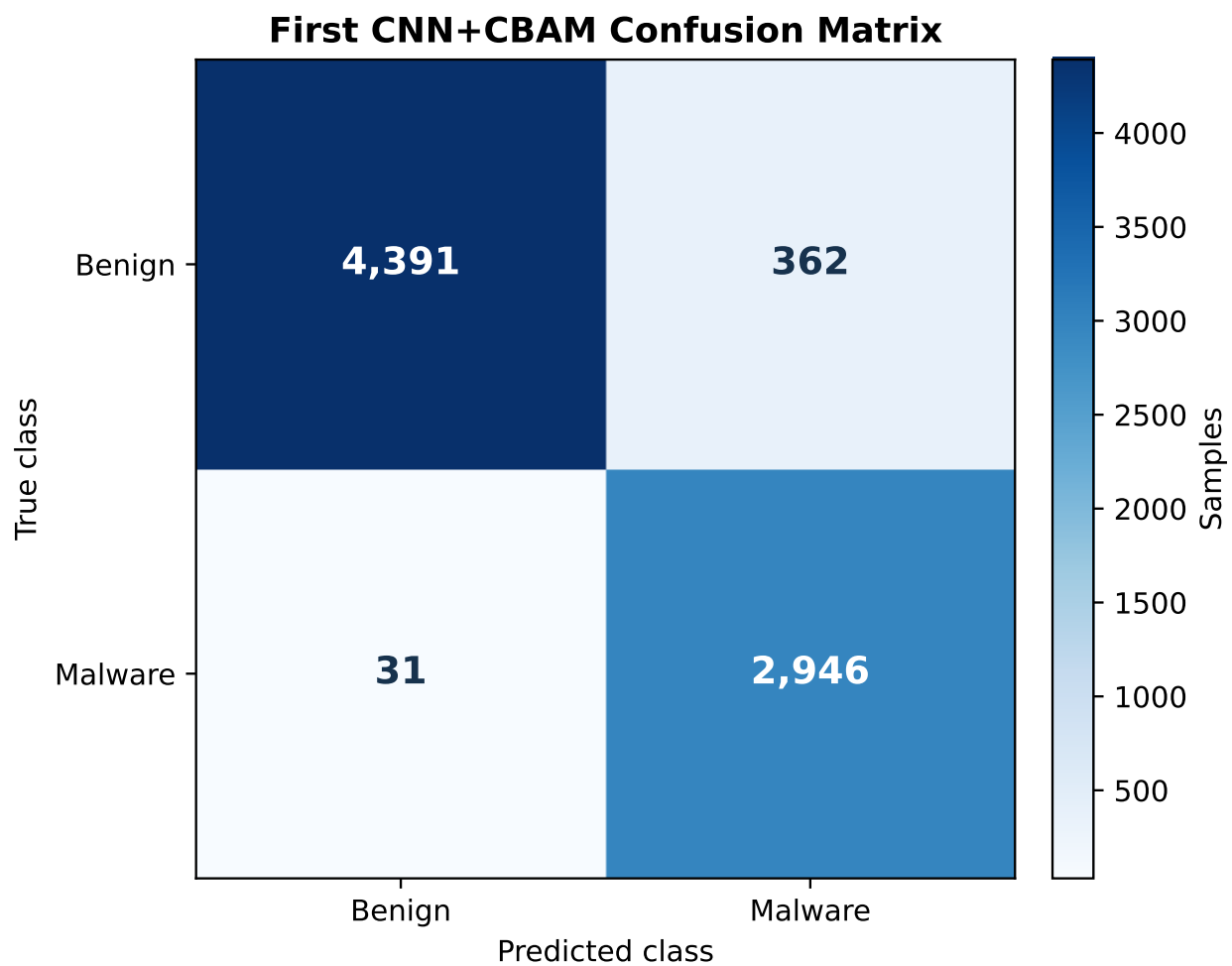


Figure 3: CNN+CBAM confusion matrix on the shared held-out test set.

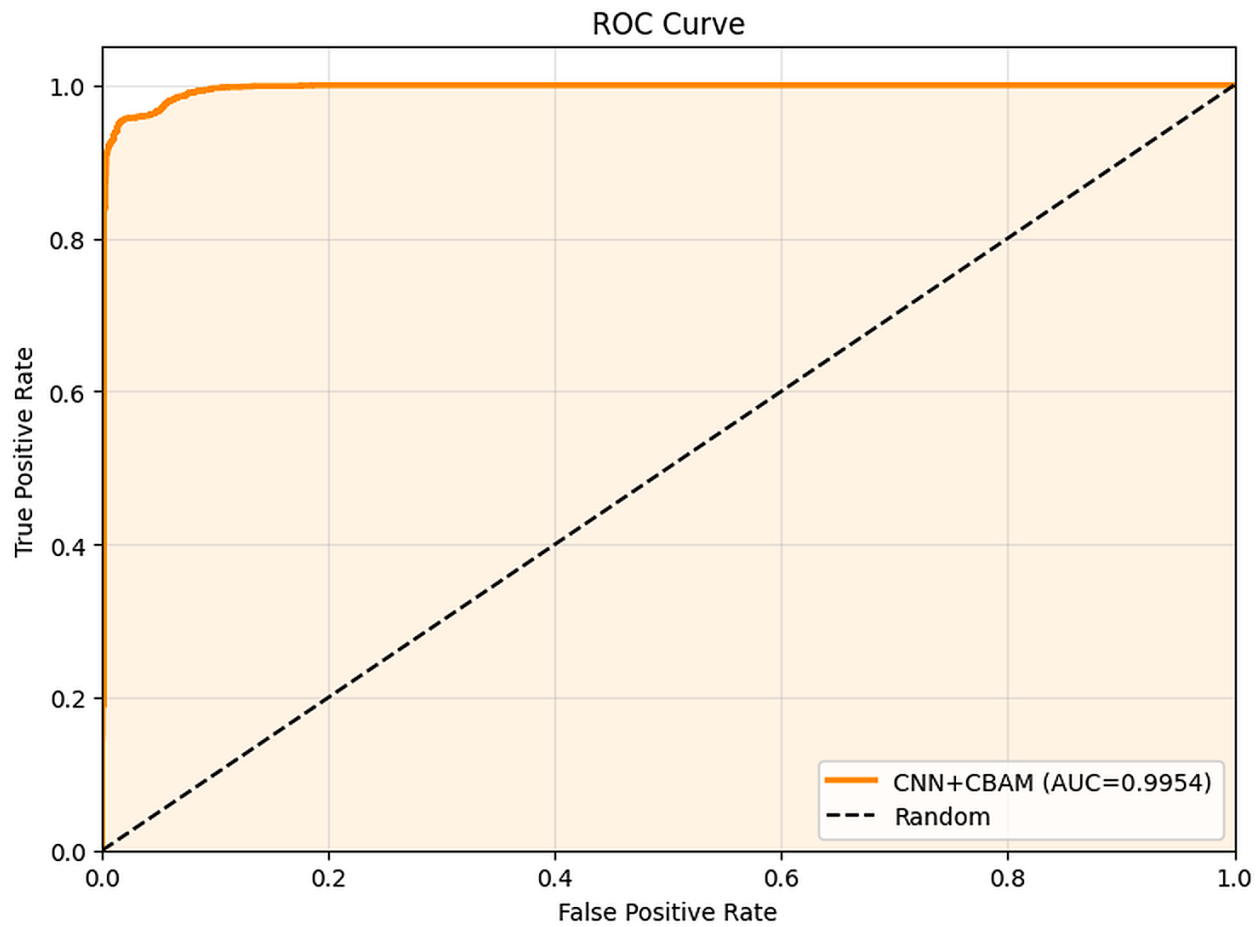


Figure 4: ROC curves for the first CNN+CBAM experiment.

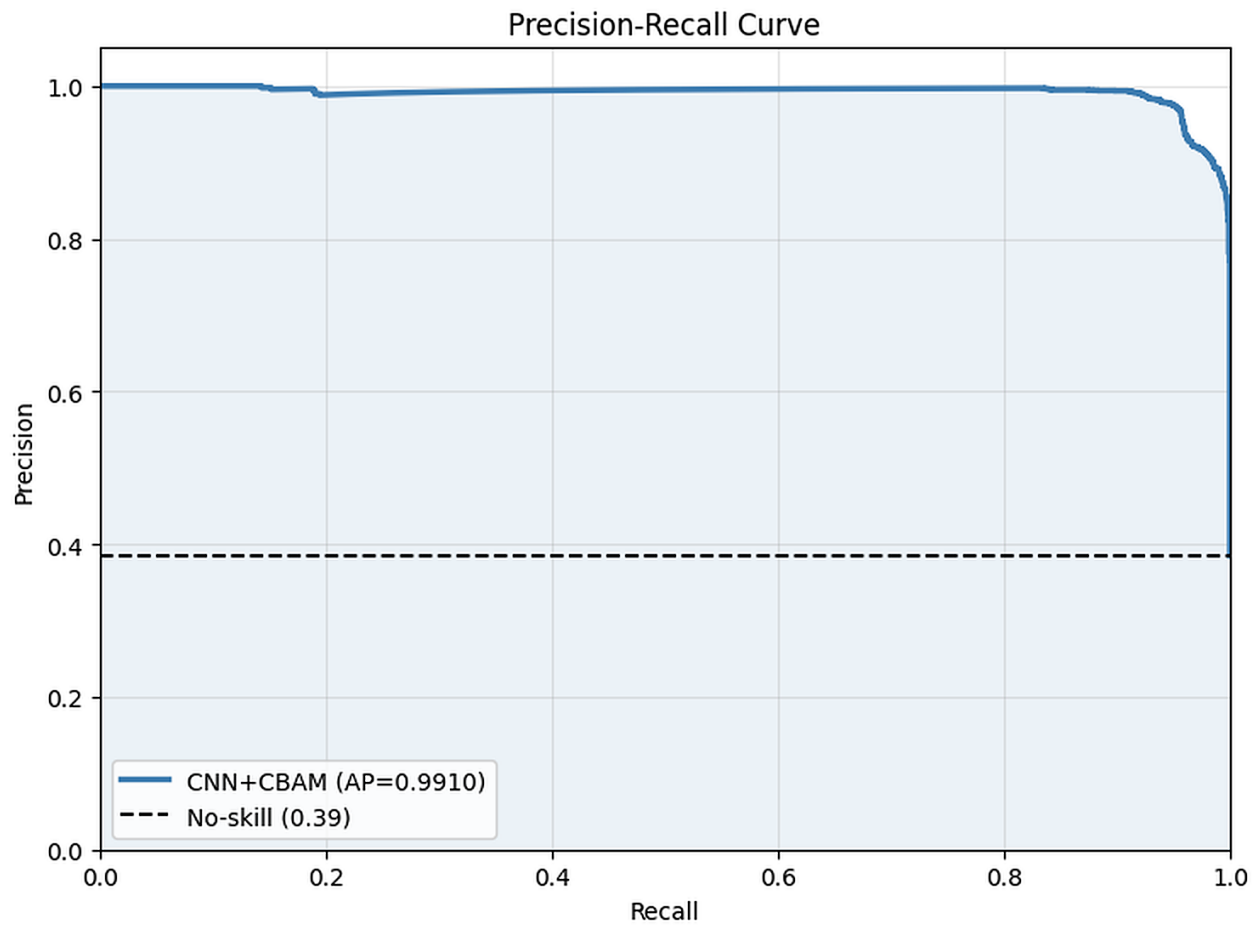


Figure 5: Precision-recall curves for the first CNN+CBAM experiment.

9. Computational comparison

The custom networks are markedly smaller than the transfer-learning baselines: 5.70 MB for the plain CNN and 5.78 MB for CNN+CBAM, compared with 57.73 MB for VGG16 and 96.62 MB for ResNet50. In the recorded runs, CBAM trained in 1,071.9 seconds and the plain CNN in 1,213.3 seconds. Since early stopping selected different epochs, this wall-clock difference cannot be attributed solely to architecture.

10. Limitations

Task 2 uses one grouped split and one random seed. The Macro-F1 difference between the custom models is below one tenth of a percentage point, so the run does not establish superiority for either architecture. The imposed feature ordering may favor particular convolutional filters, and the transfer-learning backbones were developed for natural imagery rather than repeated grayscale feature planes. Task 3 consequently adds matched dropout controls, grouped cross-validation and paired significance tests; no external validation cohort is available.

References

- [1] C. Wang, Z. Zhao, F. Wang, and Q. Li, 'MSAAM: A Multiscale Adaptive Attention Module for IoT Malware Detection and Family Classification,' *Security and Communication Networks*, 2022, doi: 10.1155/2022/2206917.
- [2] S. J. Kattamuri, R. K. V. Penmatsa, S. Chakravarty, and V. S. P. Madabathula, 'Swarm Optimization and Machine Learning Applied to PE Malware Detection towards Cyber Threat Intelligence,' *Electronics*, vol. 12, no. 2, art. 342, 2023, doi: 10.3390/electronics12020342.
- [3] V. Ravi and M. Alazab, 'Attention-based convolutional neural network deep learning approach for robust malware classification,' *Computational Intelligence*, 2023, doi: 10.1111/coin.12551.
- [4] H. Huang, R. Du, Z. Wang, et al., 'A Malicious Code Detection Method Based on Stacked Depthwise Separable Convolutions and Attention Mechanism,' *Sensors*, vol. 23, no. 16, art. 7084, 2023, doi: 10.3390/s23167084.
- [5] Y. Fan, K. Zhang, B. Zheng, et al., 'GCSA-ResNet: a deep neural network architecture for Malware detection,' *Scientific Reports*, 2025, doi: 10.1038/s41598-025-10561-6.
- [6] J. Liu, G. Zhou, Y. He, and X. Hou, 'Research on image malware classification method based on AlexNet transfer learning and MHSA attention mechanism,' *Proc. SPIE* 14122, 141220X, 2026, doi: 10.1117/12.3105850.
- [7] M. Alwateer et al., 'A hierarchical deep learning framework with doubly regularized loss for robust malware detection and family categorization,' *Scientific Reports*, 2026, doi: 10.1038/s41598-025-31444-w.
- [8] S. Woo, J. Park, J.-Y. Lee, and I. S. Kweon, 'CBAM: Convolutional Block Attention Module,' in *Proc. ECCV*, 2018, pp. 3-19.
- [9] R. R. Selvaraju et al., 'Grad-CAM: Visual Explanations from Deep Networks via Gradient-Based Localization,' in *Proc. ICCV*, 2017, pp. 618-626.
- [10] M. T. Ribeiro, S. Singh, and C. Guestrin, 'Why Should I Trust You?: Explaining the Predictions of Any Classifier,' in *Proc. KDD*, 2016, pp. 1135-1144.